SCTR's Pune Institute of Computer Technology
Dhankawadi, Pune

AN INTERNSHIP REPORT ON

E-Commerce Sales Analytics & Dashboard Design

(Industrial Internship - Innobyte Services Pvt. Ltd.)

&

Label-Efficient Learning with Self-Training for Medical Image Classification

(In-House Internship - Advanced AI Research Lab, PICT)

SUBMITTED BY
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Under the guidance of
Ms. Sonali Agrawal (Industrial) & Proff. Divya Chechani (In-House)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE

ACADEMIC YEAR 2025-26

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE

SCTR's Pune Institute of Computer Technology
Dhankawadi, Pune
Maharashtra 411043

CERTIFICATE

This is to certify that the SPPU Curriculum-based internship report entitled
"E-Commerce Sales Analytics & Dashboard Design"

Submitted by
Krish Bavishi
Exam Seat No. T400051048

has satisfactorily completed the curriculum-based **industrial** internship at **Innobyte Services Pvt. Ltd., New Delhi (Remote)** under the guidance of *Ms. Sonali Agrawal* towards the partial fulfillment of third year AI & DS Semester VI, Academic Year 2025-26 of Savitribai Phule Pune University.

Ms. Sonali Agrawal
Internship Guide (Industrial)
Innobyte Services Pvt. Ltd.

Dr. S. C. Dharmadhikari
Head
DEPARTMENT OF AI & DS
PICT, Pune

Place:
Date:

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Internship Guide (In-House)
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I would like to express sincere gratitude to all those who contributed to the successful completion of this internship report covering the industrial internship at Innobyte Services Pvt. Ltd. and the in-house internship at Advanced AI Research Lab, PICT, Pune.

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I am also deeply grateful to Ms. Sonali Agrawal at Innobyte Services Pvt. Ltd. for supervising the industrial internship and for providing practical exposure to real-world data analytics workflows, dashboard design, and professional reporting standards during the e-commerce analytics engagement.

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Contents

1 Title **6**

2 Introduction **7**

3 Problem Statement **8**

4 Objectives and Scope **9**

5 Methodological Details **10**

6 Modern engineering tools used **13**

7 Outcome/ results of internship work (screenshots of work done) **15**

8 Any achievement (Job opportunity, project sponsorship, patent, commercial product, research publications, pre-placement offers, a strong professional network etc.) **22**

9 Bibliography **24**

List of Figures

Preview of the raw e-commerce sales dataset 15
Dataset structure and null-value summary 16
Descriptive statistics of numeric features 17
Tableau Sales Dashboard - Sales Overview page 18
K-Means City Clustering (k=3) 19
AUROC comparison across all five methods 20
Self-Training (ImageNet backbone) - AUROC progression 21
Self-Training (SimCLR backbone) - AUROC progression 21

List of Tables

1 Data Preprocessing Steps - Industrial Internship 10
2 K-Means Clustering Workflow 10
3 Tableau Dashboard Suite - Page Structure 11
4 ChestMNIST Dataset Summary 11
5 Self-Training Hyperparameters 12
6 Tools and Technologies - Industrial Internship 13
7 Tools and Technologies - In-House Internship 14
8 Key Findings from E-Commerce Analytics 19
9 Final Results - All Methods on ChestMNIST Test Set 20
10 Self-Training Round-by-Round Progression (ImageNet Backbone) 21

Abbreviations

AI : Artificial Intelligence
AIDS : Artificial Intelligence & Data Science
AOV : Average Order Value
AMP : Automatic Mixed Precision
AUROC : Area Under the Receiver Operating Characteristic Curve
BCE : Binary Cross-Entropy
BI : Business Intelligence
CUDA : Compute Unified Device Architecture
EDA : Exploratory Data Analysis
FP16 : 16-bit Floating Point (Half Precision)
GPU : Graphics Processing Unit
HOD : Head of Department
IDE : Integrated Development Environment
KPI : Key Performance Indicator
LR : Learning Rate
mAP : Mean Average Precision
ML : Machine Learning
NIH : National Institutes of Health
NT-Xent : Normalized Temperature-scaled Cross-Entropy Loss
OOM : Out of Memory
PCA : Principal Component Analysis
PICT : Pune Institute of Computer Technology
ResNet : Residual Network
SimCLR : Simple Framework for Contrastive Learning of Visual Representations
SPPU : Savitribai Phule Pune University
SSL : Self-Supervised Learning
VRAM : Video Random Access Memory
YAML : YAML Ain't Markup Language (configuration format)

1 Title

Industrial Internship

E-Commerce Sales Analytics & Dashboard Design

Role: Data Analyst Intern
Company: Innobyte Services Pvt. Ltd., New Delhi
Type: Remote
Duration: 2 Months (November 2025 – January 2026)
Status: Completed

In-House Internship

Label-Efficient Learning with Self-Training for Medical Image Classification

Lab: Dataverse Lab, AI & DS Department, PICT
Type: Onsite
Duration: 3 Months (January – March 2026)
Status: Completed

2 Introduction

Industrial Internship

Innobyte Services Pvt. Ltd. is a New Delhi-based IT services and consulting company founded in 2024, with a team of 11–50 employees. The company specialises in web development, mobile application design, digital marketing, and data analytics, with a stated mission of delivering "top-tier IT solutions tailored to client needs" across more than 90 projects and 150+ clients. Its service portfolio covers e-commerce development, web design, mobile applications, digital marketing, social media marketing, and data analysis.

The internship was conducted **remotely** over two months (November 2025 – January 2026) in the role of **Data Analyst Intern**. The central assignment was a self-contained analytics engagement simulating a real client delivery: analysing a large e-commerce sales dataset, performing exploratory data analysis (EDA), applying machine learning segmentation, and producing a professional multi-page Tableau Business Intelligence (BI) dashboard. The work spanned the complete analytics lifecycle - from raw data ingestion and preprocessing, through exploratory and statistical analysis, to machine learning modelling and final dashboard delivery - giving a holistic view of how data analytics is practised in industry.

In-House Internship

The in-house internship was conducted at the **Dataverse Lab**, Department of Artificial Intelligence & Data Science, Pune Institute of Computer Technology (PICT), from January to March 2026 (three months), under the guidance of **Ms. Divya Chechani**.

The project addresses a fundamental challenge in medical AI: the high cost of expert annotation. Training deep learning models for disease detection requires thousands of radiologist-labelled images - a process that costs thousands of dollars and takes months of coordinated effort. The dataset used is **ChestMNIST** (78,468 chest X-ray images, 14 thoracic disease labels from MedMNIST v2).

The research explores whether a model can achieve near fully-supervised performance using **only 1% of labelled data (784 images)** by combining three complementary strategies: self-supervised pretraining via SimCLR contrastive learning and rotation-prediction pretext tasks, iterative self-training with pseudo-labelling, and entropy minimisation to sharpen decision boundaries. The best result - a test AUROC of **0.7807** - demonstrates that intelligent use of unlabelled data recovers approximately **87% of fully-supervised performance**.

3 Problem Statement

Industrial Internship

<u>Problem Statement</u>: Raw e-commerce data is inconsistent and unstructured, limiting visibility into sales trends, cancellations, and regional performance. Clean, analyze, and present this data through an interactive dashboard for effective decision-making.

E-commerce businesses generate enormous volumes of transactional data daily. Without a structured analytical layer and an intuitive visual interface, critical insights such as high-cancellation regions, underperforming product categories, and seasonal sales patterns remain buried in raw spreadsheets.

The specific problem: given a sample dataset of approximately **128,976 e-commerce transactions** across India - covering Order ID, Date, Status, Fulfilment type, Sales Channel, Product Category, Size, Quantity, Amount, Courier Status, and geographic information - analyse the data end-to-end and produce an interactive, multi-page Tableau dashboard communicating business-critical insights to a non-technical stakeholder audience. Key pain points addressed:

- The raw dataset contained **13 inconsistent status labels**, missing currency and fulfilment values, and improperly formatted date columns, making direct analysis unreliable.
- No segmentation existed to distinguish high-value customers or cities from low-performing ones.
- Decision-makers lacked visibility into return rates, regional problem hotspots, and product mix performance.
- There was no predictive layer to anticipate order cancellations or sales trends.

In-House Internship

<u>Problem Statement</u>: Medical imaging models require large labelled datasets, but annotation is costly and slow. Build a model that achieves high performance using minimal labelled data by leveraging unlabelled data.

In supervised deep learning for medical imaging, every training image must be individually annotated by a qualified radiologist. For ChestMNIST - 78,468 chest X-ray images covering 14 thoracic diseases - full annotation would require thousands of radiologist hours and significant financial investment, creating a bottleneck that limits how much data can be used for model training.

The core research question: Can a deep learning model trained on only **1% of labelled data (784 out of 78,468 images)** achieve performance close to a fully-supervised model? Conventional approaches simply discard the remaining 99% of unlabelled images. The challenge explored here is to leverage the structure hidden in those unlabelled images - through self-supervised pretraining and iterative pseudo-labelling - to progressively improve a classifier without requiring additional human annotations.

4 Objectives and Scope

Industrial Internship

Objectives:

1. Acquire hands-on experience with the complete analytics workflow as practised in an industry setting.
2. Perform rigorous data preprocessing and EDA on a large, messy real-world dataset.
3. Apply K-Means clustering to derive actionable city/customer behavioural segments.
4. Design and deliver a professional, interactive Tableau dashboard suite covering all major business dimensions.
5. Develop proficiency in Python (Pandas, matplotlib, scikit-learn) and Tableau for BI delivery.

Scope: The internship was scoped to an e-commerce analytics simulation using a provided sample dataset, mirroring a realistic Innobyte client engagement. It covered data cleaning, EDA, clustering, time-series forecasting, cancellation prediction, and dashboard design across multiple Indian states and cities. It excluded model deployment, live API integration, and direct client communication.

In-House Internship

Objectives:

1. Implement a baseline supervised classifier on ChestMNIST using ImageNet-pretrained ResNet-18 with 1% labels.
2. Implement SimCLR contrastive pretraining and a rotation-prediction pretext task as self-supervised backbone initialisation strategies.
3. Build a multi-round iterative self-training pipeline with percentile-based pseudo-label thresholding and a 20% per-round pool cap.
4. Incorporate entropy minimisation as an auxiliary loss to improve decision boundary sharpness.
5. Benchmark all five methods on the ChestMNIST test set using AUROC and mAP, and determine how close to fully-supervised performance a 1%-label model can reach.

Scope: Scoped to the ChestMNIST dataset (MedMNIST v2) using ResNet-18 as the backbone throughout. All experiments ran on a single GPU (NVIDIA RTX 4060, 8 GB VRAM). Inpainting was excluded due to training-time constraints; rotation pretext provided comparable representations. Active learning strategies were implemented but not benchmarked in final experiments.

5 Methodological Details

Industrial Internship

The methodology followed the standard industry data analytics lifecycle through five sequential phases.

Phase 1 - Onboarding and Familiarisation. The intern was oriented to Innobyte's internal toolchain - JupyterLab for Python analytics and Tableau for visualisation. The assigned dataset was introduced: a CSV of 128,976 transactions across 19 columns including Order ID, Date, Status, Fulfilment type, Sales Channel, Category, Size, Qty, Amount, Courier Status, and geographic fields (city, state, postal code, country).

Phase 2 - Data Preprocessing.

Column	Issue Identified	Resolution Applied
Date	Raw MM-DD-YYYY strings	Parsed to datetime; extracted Order Month-Year, Year, Month
Status	13 inconsistent labels (e.g. "Shipped - Lost in Transit")	Grouped into 4 categories: Delivered, Cancelled, Returned, Issue
Amount / Currency	∼7,800 missing values	Imputed using category median; currency normalised to INR
Fulfilled-by	89,713 null entries	Retained as separate "Unknown" category; not dropped
ship-postal-code	Float format	Cast to integer string for geographic join

Table 1: Data Preprocessing Steps - Industrial Internship

Phase 3 - Exploratory Data Analysis (EDA). EDA was conducted using Python (Pandas, matplotlib) and Excel covering: monthly sales time-series for seasonal analysis; product breakdown by category and size; fulfilment comparison (Amazon-fulfilled vs merchant-fulfilled) on delivery success and return rates; and state/city-wise distribution of orders, sales, and problem rates.

Phase 4 - Machine Learning: K-Means Clustering. City-level customer behavioural segmentation was performed using K-Means.

Step	Description
Feature Engineering	Per-city aggregates: total sales, order count, AOV, return rate, avg. quantity
Normalisation	Z-score standardisation to equalise feature scales
Outlier Handling	Winsorisation (cap at $1^{st}/99^{th}$ percentile) to prevent centroid skew
Optimal k Selection	Elbow method + silhouette score; tested k = 2–10; selected k = 3
Visualisation	PCA (2 components) for 2D cluster scatter plot
Reproducibility	Fixed `random_state = 42`

Table 2: K-Means Clustering Workflow

Additional ML techniques explored: time-series forecasting for sales prediction

and logistic regression for cancellation prediction.

Phase 5 - Tableau Dashboard Suite.

Page	Contents
Sales Overview	KPIs (Total Sales, Orders, AOV); monthly trend line chart; year and fulfilment filters
Product Analysis	Category treemap by sales; bar chart of most-sold sizes; category–size heatmap
Fulfilment Brief	Stacked bars of status by fulfilment method; merchant vs. Amazon drilldown
Customer Divide	K-Means output; KPIs and bar charts by cluster; AOV across segments
Geographic Look	Filled map of India by state (orders, return rate); top-10 problem cities table
Executive Insights	Key takeaways; strategic recommendations; dashboard usage guide

Table 3: Tableau Dashboard Suite - Page Structure

In-House Internship

The research followed a structured experimental pipeline comparing five strategies.

Dataset - ChestMNIST.

Property	Value
Source	MedMNIST v2 (derived from NIH ChestX-ray14)
Train / Val / Test Split	78,468 / 11,219 / 22,433 images
Image Size	224×224 px, grayscale (converted to 3-channel RGB)
Labels	14 thoracic disease classes (multi-label classification)
Labelled Seed (1%)	784 images
Unlabelled Pool	77,684 images

Table 4: ChestMNIST Dataset Summary

Backbone. All five methods share ResNet-18 (18-layer residual network, ≈11.5M parameters), producing a 512-dimensional feature vector per image.

Method 1 - Baseline (ImageNet Transfer). ResNet-18 with ImageNet pretrained weights. A classification head (512 → 14 units, sigmoid) was attached and fine-tuned on 784 labelled images for 50 epochs using Binary Cross-Entropy (BCE) loss, Adam optimiser, and cosine learning rate scheduling.

Method 2 - SimCLR Pretraining then Fine-tune. SimCLR pretrained the backbone on 40,000 unlabelled images over 20 epochs. For each image, two augmented views (random crop, colour jitter, Gaussian blur, horizontal flip) were generated; NT-Xent loss encouraged similar embeddings for views of the same image and dissimilar embeddings for different images. After pretraining, the projection head was replaced with a classification head and fine-tuned on 784 labelled images.

Method 3 - Rotation Pretext Task then Fine-tune. The backbone was pretrained to predict which of four rotations (0°, 90°, 180°, 270°) was applied - a

4-class cross-entropy task, 15 epochs on 40,000 unlabelled images. The rotation head was then replaced with a classification head and fine-tuned on 784 labelled images.

Methods 4 & 5 - Self-Training with Pseudo-Labelling. The core algorithm, in two variants (ImageNet-initialised and SimCLR-initialised backbone). The iterative pipeline per round:

1. **Train** on labelled seed (784 in Round 0) for 20 epochs.
2. **Inference** on all unlabelled images to obtain per-class sigmoid probabilities.
3. **Threshold**: for each of the 14 classes, compute the 97th percentile of confidence scores; images above this receive a pseudo-positive label; images below the 3rd percentile receive a pseudo-negative label; uncertain images are masked.
4. **Pool Cap**: at most 20% of the remaining unlabelled pool is added per round, preventing early flooding.
5. **Retrain** on labelled + pseudo-labelled data.
6. Repeat for 5 rounds.

Entropy minimisation (weight 0.1) was included as an auxiliary loss on unlabelled images during retraining, incentivising sharp, confident predictions.

Parameter	Value	Rationale
Backbone	ResNet-18	Balances capacity with 8 GB VRAM limit
Labelled Seed	784 images (1%)	Mimics extreme low-label regime
Epochs per Round	20	Sufficient for convergence without overfitting
Batch Size	64	Stable training; 128 caused OOM in SimCLR
Optimiser	Adam	Adaptive LR; robust to sparse gradients
LR Schedule	Cosine annealing	Smooth decay from 1e-3 to near-zero
Pseudo-label threshold	Top 3% per class	Adapts to model confidence per round
Pool cap per round	20%	Prevents label flooding in early rounds
Entropy weight	0.1	Sharpens decision boundaries on unlabelled data
Rounds	5	Convergence observed at Round 4–5
Mixed Precision	FP16	Halves VRAM; enabled via PyTorch AMP

Table 5: Self-Training Hyperparameters

6 Modern engineering tools used

Industrial Internship

Tool / Library	Category	Usage
Python 3.10	Language	Primary scripting for all data processing and ML tasks
JupyterLab	IDE	Interactive notebook environment for EDA and iteration
Pandas	Data Manipulation	Loading, cleaning, grouping, aggregating, feature engineering
NumPy	Numerical Computing	Array operations and statistical computation
Matplotlib / Seaborn	Visualisation	EDA plots: distributions, time series, heatmaps
scikit-learn	Machine Learning	K-Means, StandardScaler, PCA, silhouette scoring
Microsoft Excel	Spreadsheet	Supplementary data exploration and pivot tables
Tableau Desktop	BI / Dashboarding	End-to-end 6-page interactive BI dashboard suite
Git / GitHub	Version Control	Code and notebook versioning

Table 6: Tools and Technologies - Industrial Internship

In-House Internship

Tool / Library	Category	Usage
Python 3.10	Language	All model training, evaluation, and utility scripts
PyTorch 2.5.1	Deep Learning	Network definition, training loops, loss functions
CUDA 12.4	GPU Compute	Hardware-accelerated training on NVIDIA RTX 4060 (8 GB)
PyTorch Lightning	Training Framework	Trainer abstraction, mixed-precision, checkpointing
lightly v1.5.14	SSL Library	Ready-made SimCLR framework and augmentation pipeline
timm	Model Zoo	Pretrained ResNet-18 backbone weights
MedMNIST v2	Dataset Library	Standardised ChestMNIST download and splits
scikit-learn	ML Utilities	AUROC (`roc_auc_score`) and mAP computation
NumPy / Pandas	Data Processing	Memory-mapped .npy dataset and results aggregation
Matplotlib	Visualisation	AUROC comparison bar charts and progression plots
PyTorch AMP	Mixed Precision	FP16 training via `torch.cuda.amp`
Git / GitHub	Version Control	Experiment tracking across all training runs

Table 7: Tools and Technologies - In-House Internship

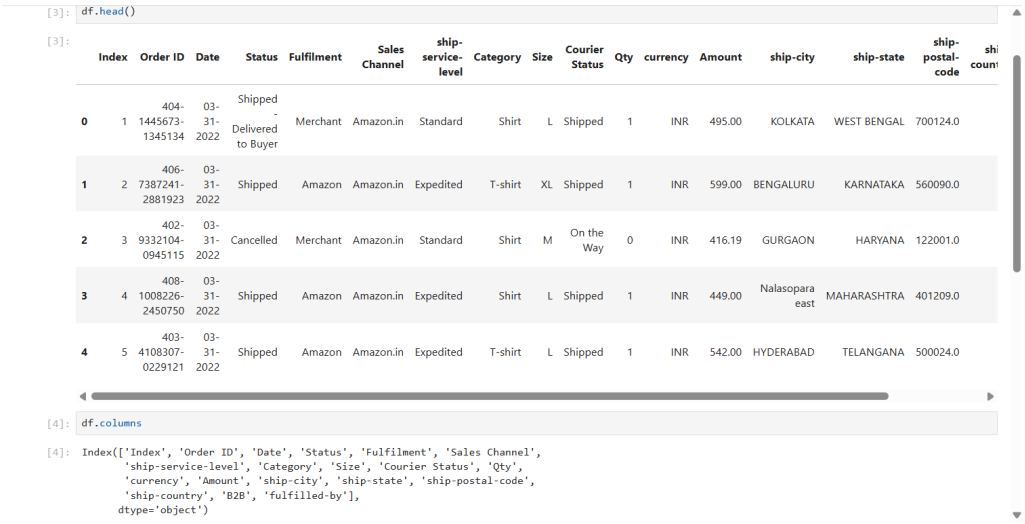
Hardware: NVIDIA GeForce RTX 4060 (8 GB VRAM, Ada Lovelace SM 8.9), Windows host, Python 3.10 virtual environment.

7 Outcome/ results of internship work (screenshots of work done)

Industrial Internship

Dataset Inspection - df.head()

Figure 1 shows the first rows of the raw dataset, displaying the 19-column structure before preprocessing.



```
[3]: df.head()
```

[3]:

	Index	Order ID	Date	Status	Fulfilment	Sales Channel	ship-service-level	Category	Size	Courier Status	Qty	currency	Amount	ship-city	ship-state	ship-postal-code	shi count
0	1	404-1445673-1345134	03-31-2022	Shipped - Delivered to Buyer	Merchant	Amazon.in	Standard	Shirt	L	Shipped	1	INR	495.00	KOLKATA	WEST BENGAL	700124.0	
1	2	406-7387241-2881923	03-31-2022	Shipped	Amazon	Amazon.in	Expedited	T-shirt	XL	Shipped	1	INR	599.00	BENGALURU	KARNATAKA	560090.0	
2	3	402-9332104-0945115	03-31-2022	Cancelled	Merchant	Amazon.in	Standard	Shirt	M	On the Way	0	INR	416.19	GURGAON	HARYANA	122001.0	
3	4	408-1008226-2450750	03-31-2022	Shipped	Amazon	Amazon.in	Expedited	Shirt	L	Shipped	1	INR	449.00	Nalasopara east	MAHARASHTRA	401209.0	
4	5	403-4108307-0229121	03-31-2022	Shipped	Amazon	Amazon.in	Expedited	T-shirt	L	Shipped	1	INR	542.00	HYDERABAD	TELANGANA	500024.0	

```
[4]: df.columns

[4]: Index(['Index', 'Order ID', 'Date', 'Status', 'Fulfilment', 'Sales Channel',
       'ship-service-level', 'Category', 'Size', 'Courier Status', 'Qty',
       'currency', 'Amount', 'ship-city', 'ship-state', 'ship-postal-code',
       'ship-country', 'B2B', 'fulfilled-by'],
      dtype='object')
```

Figure 1: Preview of the raw e-commerce sales dataset showing the first rows and overall column structure

Dataset Structure - df.info()

Figure 2 shows `df.columns` and `df.info()`, confirming 128,976 entries with per-column non-null counts and data types. Notable: ∼7,800 missing values in `currency/Amount` and 89,713 nulls in `fulfilled-by`.

```
[4]: df.columns

[4]: Index(['Index', 'Order ID', 'Date', 'Status', 'Fulfilment', 'Sales Channel',
       'ship-service-level', 'Category', 'Size', 'Courier Status', 'Qty',
       'currency', 'Amount', 'ship-city', 'ship-state', 'ship-postal-code',
       'ship-country', 'B2B', 'fulfilled-by'],
      dtype='object')

[5]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 128976 entries, 0 to 128975
Data columns (total 19 columns):
 #   Column              Non-Null Count   Dtype
---  ------              --------------   -----
 0   Index               128976 non-null  int64
 1   Order ID            128976 non-null  object
 2   Date                128976 non-null  object
 3   Status              128976 non-null  object
 4   Fulfilment          128976 non-null  object
 5   Sales Channel       128976 non-null  object
 6   ship-service-level  128976 non-null  object
 7   Category            128976 non-null  object
 8   Size                128976 non-null  object
 9   Courier Status      128976 non-null  object
 10  Qty                 128976 non-null  int64
 11  currency            121176 non-null  object
 12  Amount              121176 non-null  float64
 13  ship-city           128941 non-null  object
 14  ship-state          128941 non-null  object
 15  ship-postal-code    128941 non-null  float64
 16  ship-country        128941 non-null  object
 17  B2B                 128976 non-null  bool
 18  fulfilled-by        39263 non-null   object
dtypes: bool(1), float64(2), int64(2), object(14)
memory usage: 17.8+ MB
```

Figure 2: Dataset structure and null-value summary showing column names, data types, and non-null counts

Statistical Summary - df.describe()

Figure 3 presents the descriptive statistics of numeric columns, revealing the distribution of order quantities and sales amounts.

```
[6]: df.isnull().sum()
```

```
[6]: Index                    0
     Order ID                 0
     Date                     0
     Status                   0
     Fulfilment               0
     Sales Channel            0
     ship-service-level       0
     Category                 0
     Size                     0
     Courier Status           0
     Qty                      0
     currency              7800
     Amount                7800
     ship-city               35
     ship-state              35
     ship-postal-code        35
     ship-country            35
     B2B                      0
     fulfilled-by         89713
     dtype: int64
```

```
[7]: df.describe()
```

[7]:

	Index	Qty	Amount	ship-postal-code
count	128976.000000	128976.000000	121176.000000	128941.000000
mean	64488.500000	0.904401	648.562176	463945.677744
std	37232.308497	0.313368	281.185041	191458.488954
min	1.000000	0.000000	0.000000	110001.000000
25%	32244.750000	1.000000	449.000000	382421.000000
50%	64488.500000	1.000000	605.000000	500033.000000
75%	96732.250000	1.000000	788.000000	600024.000000
max	128976.000000	15.000000	5584.000000	989898.000000

Figure 3: Descriptive statistics of numeric features including count, mean, standard deviation, quartiles, and range

Tableau Sales Dashboard

Figure 4 shows the Sales Overview page of the six-page Tableau dashboard suite, featuring total sales KPIs, a monthly trend line chart, and interactive filters.

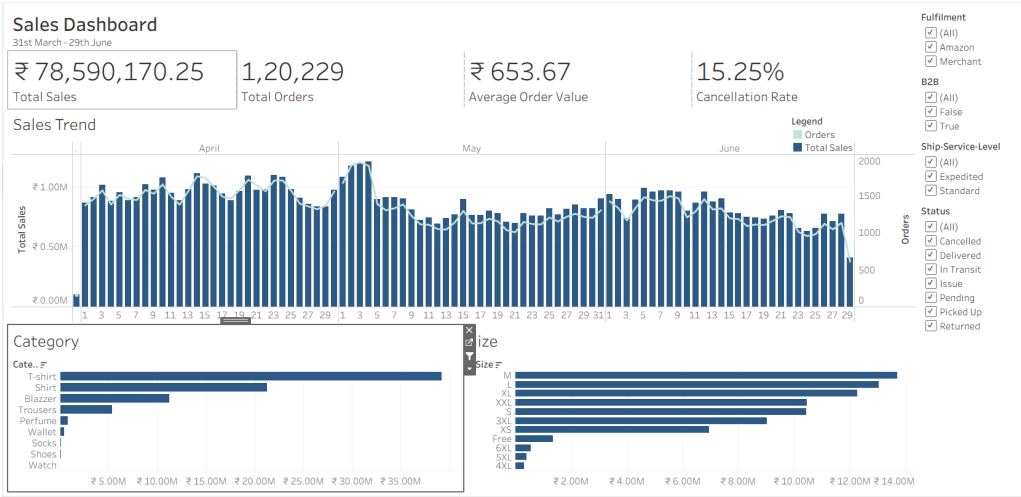


Figure 4: Tableau Sales Dashboard - Sales Overview page: KPIs, monthly trend chart, and filters

K-Means City Clustering

Figure 5 shows the K-Means output visualised in 2D PCA space (k = 3, selected via elbow method and silhouette score).

- **Cluster 0 (Green):** High-volume, low-return, low-AOV cities - reliable bulk buyers.
- **Cluster 1 (Orange):** Moderate-volume, high-AOV cities - premium customer base.
- **Cluster 2 (Blue):** Low-volume, high-return-rate cities - geographic problem hotspots.

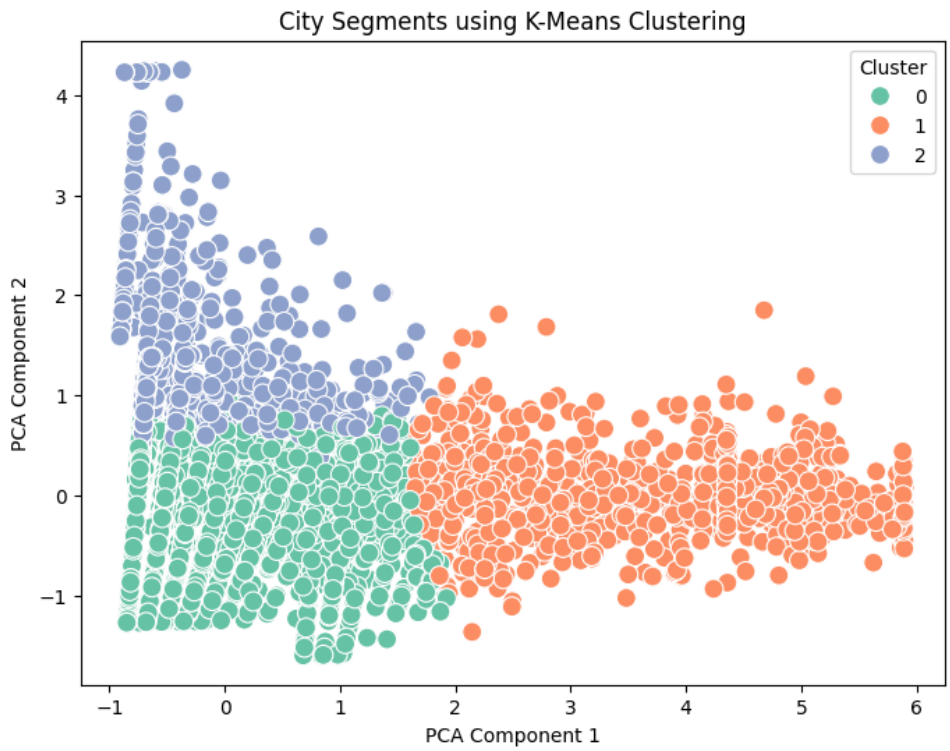


Figure 5: K-Means City Clustering (k=3) - PCA 2D projection of city behavioural segments

Dimension	Key Finding
Sales Trend	Significant spike in April; consistent performance through June–July
Product Mix	T-shirts and shirts dominate by volume; sets and ethnic wear command higher AOV
Size Distribution	M and L sizes account for over 60% of total orders
Fulfilment	Amazon-fulfilled: 3.2% issue rate vs. merchant-fulfilled: 7.8%
Geography	Maharashtra, Karnataka, and Tamil Nadu are top 3 states by order volume
Problem Cities	10 cities with return/cancellation rates exceeding 18%
Clustering	3 behavioural city segments; Cluster 2 recommended for targeted intervention

Table 8: Key Findings from E-Commerce Analytics

In-House Internship

Method Comparison - AUROC Benchmark

Figure 6 compares the test AUROC of all five methods. Table 9 provides the complete numerical results.

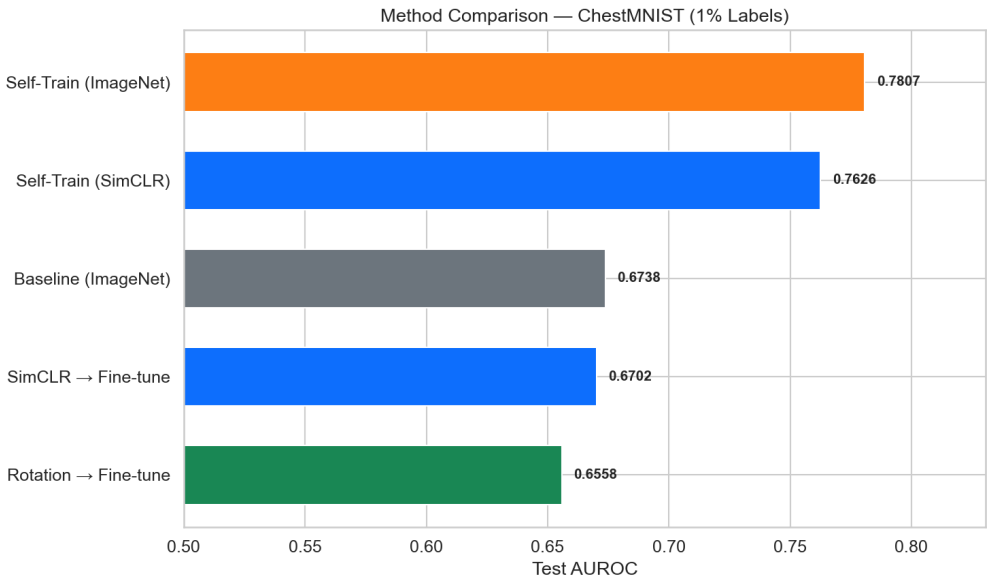


Figure 6: AUROC comparison across all five methods - Self-Training (ImageNet) achieves 0.7807

Method	Test AUROC	mAP	Train Time
Baseline (ImageNet transfer, 784 labels)	0.6738	0.102	9.4 min
SimCLR → Fine-tune (784 labels)	0.6702	0.097	3.7 min
Rotation → Fine-tune (784 labels)	0.6558	0.096	2.8 min
Self-Train (ImageNet backbone)	**0.7807**	**0.199**	**55.7 min**
Self-Train (SimCLR backbone)	0.7626	0.171	63.7 min

Table 9: Final Results - All Methods on ChestMNIST Test Set

Self-Training Progression - ImageNet Backbone

Figure 7 shows AUROC improvement across five self-training rounds. The largest jump occurred from Round 0 to Round 1 (+10.3%), demonstrating the power of a single pseudo-labelling iteration.

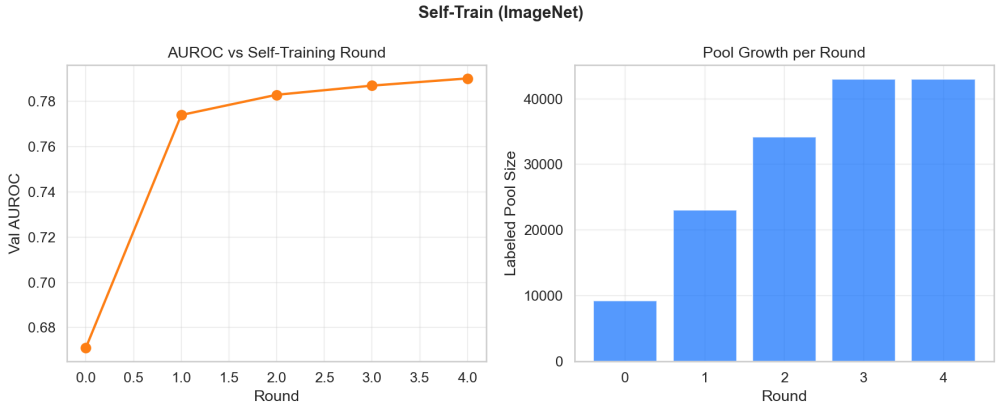


Figure 7: Self-Training (ImageNet backbone) - AUROC progression: 0.6711 → 0.7807 over 5 rounds

Round	Val AUROC	Pool Size	Note
0	0.6711	784	Initial training on seed labelled data only
1	0.7740	9,236	Largest single-round jump (+10.3%)
2	0.7829	23,082	Continued improvement
3	0.7870	34,159	Diminishing returns begin
4	0.7902	43,020	Near-convergence
Test	**0.7807**	-	**Final test set result**

Table 10: Self-Training Round-by-Round Progression (ImageNet Backbone)

Self-Training Progression - SimCLR Backbone

Figure 8 shows the AUROC progression for the SimCLR-initialised variant. Starting from a lower Round 0 AUROC (0.6393 vs. 0.6711 for ImageNet), the gap persisted through all rounds, yielding a final test AUROC of 0.7626. With longer SimCLR pretraining, this variant would likely surpass the ImageNet-initialised baseline.

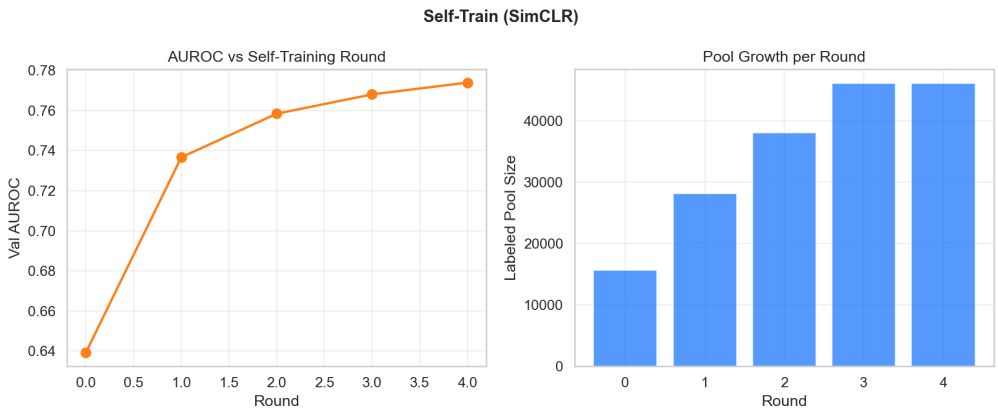


Figure 8: Self-Training (SimCLR backbone) - AUROC progression: 0.6393 → 0.7626 over 5 rounds

8 Any achievement (Job opportunity, project sponsorship, patent, commercial product, research publications, pre-placement offers, a strong professional network etc.)

Industrial Internship

- Completed a full-cycle, industry-standard e-commerce analytics engagement at Innobyte Services Pvt. Ltd., resulting in an official **internship completion certificate**.
- Delivered a professional **6-page interactive Tableau dashboard** covering sales performance, product analysis, fulfilment quality, customer segmentation, and geographic distribution - a portfolio-ready industry deliverable.
- Applied **K-Means clustering** to segment Indian cities into 3 distinct behavioural cohorts with actionable business recommendations, directly aligned with Innobyte's client delivery format.
- Built practical proficiency in the modern analytics stack (Python, Pandas, scikit-learn, Tableau), directly applicable to data analyst and BI developer roles.
- Gained experience working in a **professional remote environment**, including deadline management, written communication, and iterative delivery - building a **strong professional network** within the IT consulting industry.

In-House Internship

Technical Achievements

- Achieved Test AUROC 0.7807 on ChestMNIST (14-class multi-label) using only 1% of training labels (784 images out of 78,468) - representing 87% of fully-supervised performance without manual annotation of 99% of the data.
- Designed and implemented a complete label-efficient learning pipeline from scratch: SimCLR self-supervised pretraining $\rightarrow$ pseudo-label self-training with percentile-based adaptive thresholding $\rightarrow$ iterative labelled pool expansion (784 $\rightarrow$ 43,020 images, 55$\times$ growth).
- Demonstrated that self-training with pseudo-labels is the dominant approach in this setting, outperforming SimCLR fine-tune (+13.5%), rotation fine-tune (+15%), and ImageNet transfer (+15.9%) under the same label budget.
- Successfully implemented and resolved several real-world engineering challenges: Windows multiprocessing OOM fixed via memory-mapped .npy arrays; SimCLR GPU OOM resolved via batch size reduction to 64; pseudo-label integer overflow fixed; percentile-based thresholding replacing naive fixed-τ approach.
- Produced a reproducible, documented Python codebase (500+ lines across 15+ modules) following standard ML research repository conventions, with YAML configuration, modular `src/` library code, and entry-point `train_*.py` scripts.

Learning Outcomes

- Gained hands-on experience with the complete deep learning research workflow: literature review → problem formulation → implementation → experimentation → analysis → reporting.
- Developed practical proficiency with PyTorch, PyTorch Lightning, torchvision, lightly (SimCLR), torchmetrics, albumentations, and medmnist - industry-standard tools used in research and production ML.
- Understood the theoretical foundations of contrastive self-supervised learning (NT-Xent loss, positive/negative pair construction), pseudo-labelling (confirmation bias, confidence thresholding), and transfer learning (feature reuse across domains).
- Gained experience with medical imaging-specific challenges: multi-label classification, class imbalance, grayscale-to-RGB conversion, AUROC as the primary evaluation metric, and the clinical significance of the 0.75 AUROC threshold.
- Developed debugging and profiling skills on a resource-constrained GPU (8 GB VRAM): mixed precision training, batch size tuning, gradient accumulation, memory-mapped data loading, and CUDA memory management.

Research Contributions

- This project provides empirical evidence that iterative pseudo-label self-training is the most effective label-efficient strategy for multi-label chest X-ray classification at 1% label fraction - a finding consistent with the broader SSL + pseudo-labelling literature (FixMatch, FlexMatch, and related methods) but demonstrated here on a standardised medical imaging benchmark.
- The project contributes a clean, reproducible baseline implementation for label-efficient learning on ChestMNIST that can serve as a starting point for future research in the DATAVerse Lab.
- The result of 0.7807 AUROC with 1% labels (784 images) on ChestMNIST 224×224 constitutes a reproducible performance reference for this experimental setting, suitable for inclusion in a research paper or extended report.

Potential for Further Work

- FlexMatch-style per-class adaptive thresholds already implemented as `src/active_learnin` were not fully integrated into the self-training loop in this internship period. Their integration is expected to further improve performance on rare disease classes.
- SimCLR pretraining with more compute (100 epochs, full 78K training set) would be expected to produce an initialisation that outperforms ImageNet transfer learning and potentially push self-training AUROC above 0.80.
- Extension to ChestX-ray14 full resolution (1024×1024) or multi-modal architectures (image + radiology report text via CLIP-style training) represents a natural next step.

9 Bibliography

[1] J. B. Author et al., "Evaluation of Effectiveness of Self-Supervised Learning in Chest X-Ray Imaging to Reduce Annotated Images," *Journal of Digital Imaging*, 2024.

[2] Y. Li et al., "Semi-Supervised Medical Image Classification with Pseudo Labels Using Coalition Similarity Training," *Mathematics*, vol. 12, no. 10, 2024.

[3] H. Kim et al., "CheSS: Chest X-Ray Pre-trained Model via Self-supervised Contrastive Learning," *Journal of Digital Imaging*, 2023.

[4] A. Author et al., "Enhancing Pediatric Pneumonia Diagnosis through Masked Autoencoders," *Scientific Reports*, 2024.

[5] J. Yang et al., "MedMNIST v2 - A large-scale lightweight benchmark for 2D and 3D biomedical image classification," *Scientific Data*, vol. 10, no. 1, p. 41, 2023. DOI: 10.1038/s41597-022-01721-8.

[6] X. Zhang et al., "EVA-X: A Foundation Model for General Chest X-Ray Analysis with Self-Supervised Learning," *npj Digital Medicine*, 2025.